the white-dwarf graveyard, or (if one believed Zwicky's mechanism for supernovae, which few people did) they might eject so much matter in supernova explosions that they reduce themselves below about 1 solar mass during the explosion and wind up in the neutron-star graveyard. Most astronomers, through the 1940s and 1950s, and into the early 1960s–if they thought at all about the issue–espoused this view.

Second, besides the white-dwarf, neutron-star, and black-hole graveyards, there might be a fourth graveyard for massive stars, a graveyard unknown in the 1930s. For example, one could imagine a graveyard in Figure 5.3 at circumferences intermediate between neutron stars and white dwarfs—a few hundred or a thousand kilometers. The shrinkage of a massive star might be halted in such a graveyard before the star ever gets small enough to form either a neutron star or a black hole.

If World War II and then the cold war had not intervened, Oppenheimer and his students, or others, would likely have explored such a possibility in the 1940s and would have showed firmly that there is no such fourth graveyard.

However, World War II *did* intervene, and it absorbed the energies of almost all the world's theoretical physicists; then after the war, crash programs to develop hydrogen bombs delayed further the return of physicists to normalcy (see the next chapter).

Finally, in the mid-1950s, two physicists emerged from their respective hydrogen bomb efforts and took up where Oppenheimer and his students had left off. They were John Archibald Wheeler at Princeton University in the United States and Yakov Borisovich Zel'dovich at the Institute of Applied Mathematics in Moscow—two superb physicists, who will be major figures in the rest of this book.

## Wheeler

In March 1956, Wheeler devoted several days to studying the articles by Chandrasekhar, Landau, and Oppenheimer and Volkoff. Here, he recognized, was a mystery worth probing. Could it really be true that stars more massive than about 1.4 Suns have no choice, when they die, but to form black holes? "Of all the implications of general relativity for the structure and evolution of the Universe, this question of the fate of great